Chapter 24

Embeddings for Scientometrics

R. Heller · Date: 2026-05-09 · Last changed: 2026-10-31

24.1 Learning objectives

After completing this chapter, you will be able to:

- Explain how word and document embeddings differ from bag-of-words representations
- Train simple word embeddings on a bibliometric corpus
- Use pre-trained embeddings to compute document similarity
- Apply UMAP or t-SNE to visualise document clusters in embedding space
- Assess when embeddings add value over simpler text representations

24.2 Setup

```
library(tidyverse)
library(openalexR)
library(quanteda)
library(word2vec)
library(uwot)
library(glue)
```

```
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

24.3 Conceptual background

Bag-of-words representations (21.3) treat each word as an independent dimension. “Citation analysis” and “bibliometric study” share no features despite being semantically related. **Word embeddings** address this by mapping words into a dense, low-dimensional vector space where semantically similar words are close together. The word2vec algorithm learns these representations by predicting words from their context (skip-gram) or context from words (CBOW) in a large training corpus.

Document embeddings extend the idea to whole documents. Simple approaches average the word vectors of a document’s words (mean pooling). More sophisticated approaches include Doc2Vec (Paragraph Vector), which learns document-specific vectors alongside word vectors, and transformer-based models like **SPECTER**, which is pre-trained on scientific paper titles and abstracts using citation-based training signals.

For scientometrics, embeddings enable several applications:

- **Semantic similarity**: Finding papers with similar meaning, even when they use different terminology.
- **Clustering**: Grouping papers by content in embedding space, complementing topic models (22.3).
- **Visualisation**: Projecting embeddings to 2D with UMAP or t-SNE to create interpretable “maps” of a corpus.
- **Anomaly detection**: Identifying papers that are semantically distant from their assigned field or cluster.

The trade-off is interpretability: while TF-IDF features are directly readable (the word “citation” has a specific weight), embedding dimensions are opaque. Embeddings are powerful for similarity and clustering but resist the kind of term-level interpretation that co-word analysis provides.

24.4 Worked example

24.4.1 Preparing text data

```
works <- fetch_openalex(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2019-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 400, seed = 42)
)

text_df <- works |>
  filter(!is.na(abstract), nchar(abstract) > 100) |>
  transmute(
    doc_id = id,
    title = display_name,
    text = paste(display_name, abstract, sep = ". "),
    year = year(publication_date),
    cited_by_count
  )

cat(glue("Documents for embedding: {nrow(text_df)}\n"))
```

```
#> Documents for embedding: 108
```

24.4.2 Training word2vec embeddings

```
clean_text <- text_df$text |>
  str_to_lower() |>
  str_replace_all("[^a-z ]", " ") |>
  str_squish()

model <- word2vec(clean_text, dim = 100, iter = 20, min_count =
↪  3,
                  type = "skip-gram", threads = 1)

cat(glue("Vocabulary size: {nrow(as.matrix(model))}\n"))
```

```
#> Vocabulary size: 1395
```

```
if ("citation" %in% rownames(as.matrix(model))) {
  nn <- predict(model, newdata = "citation", type = "nearest",
 ↪  top_n = 10)
  cat("Nearest neighbours of 'citation':\n")
  print(nn)
}
```

```
#> Nearest neighbours of 'citation':
#> $citation
#>       term1         term2 similarity rank
#> 1  citation    inequality      0.719    1
#> 2  citation          mean      0.709    2
#> 3  citation    normalised      0.702    3
#> 4  citation     regarding      0.696    4
#> 5  citation        skewed      0.687    5
#> 6  citation         links      0.680    6
#> 7  citation   comparisons      0.675    7
#> 8  citation normalisation      0.667    8
#> 9  citation        source      0.660    9
#> 10 citation distributions      0.660   10
```

24.4.3 Computing document embeddings via mean pooling

Apply the same text preprocessing used for training: lowercase before removing characters outside the vocabulary alphabet. This preserves capitalised words and acronyms when matching them to the learned word vectors.

```
word_matrix <- as.matrix(model)

doc_embed <- function(text, wm) {
  words <- str_split(str_replace_all(str_to_lower(text), "[^a-z
 ↪  ]", " "),
                     "\\s+")[[1]]
  matched <- words[words %in% rownames(wm)]
  if (length(matched) == 0) return(rep(NA_real_, ncol(wm)))
  colMeans(wm[matched, , drop = FALSE])
}

doc_embeddings <- map(text_df$text, \(t) doc_embed(t,
 ↪  word_matrix))
embed_mat <- do.call(rbind, doc_embeddings)

valid <- complete.cases(embed_mat)
embed_mat <- embed_mat[valid, ]
```

```
text_df_valid <- text_df[valid, ]

cat(glue("Valid document embeddings: {nrow(embed_mat)}\n"))
```

```
#> Valid document embeddings: 108
```

24.4.4 Dimensionality reduction with UMAP

```
umap_result <- umap(embed_mat, n_neighbors = 15, min_dist = 0.1,
                    n_components = 2, ret_model = FALSE)

text_df_valid$umap_x <- umap_result[, 1]
text_df_valid$umap_y <- umap_result[, 2]
```

```
ggplot(text_df_valid, aes(x = umap_x, y = umap_y, colour =
↪  factor(year))) +
  geom_point(alpha = 0.6, size = 1.5) +
  scale_colour_manual(values =
  ↪  palette_sci(n_distinct(text_df_valid$year))) +
  labs(x = "UMAP 1", y = "UMAP 2", colour = "Year") +
  theme_sci() +
  theme(axis.text = element_blank(), axis.ticks =
  ↪  element_blank())
```

24.4.5 Document similarity search

```
cosine_sim <- function(a, b) sum(a * b) / (sqrt(sum(a^2)) *
↪  sqrt(sum(b^2)))

query_idx <- 1
sims <- map_dbl(seq_len(nrow(embed_mat)),
                \(i) cosine_sim(embed_mat[query_idx, ],
↪  embed_mat[i, ]))

top_similar <- order(sims, decreasing = TRUE)[2:6]

cat(glue("Query: {text_df_valid$title[query_idx]}\n\n"))
```

```
#> Query: Market forces in Italian academia today (and yesterday)
```

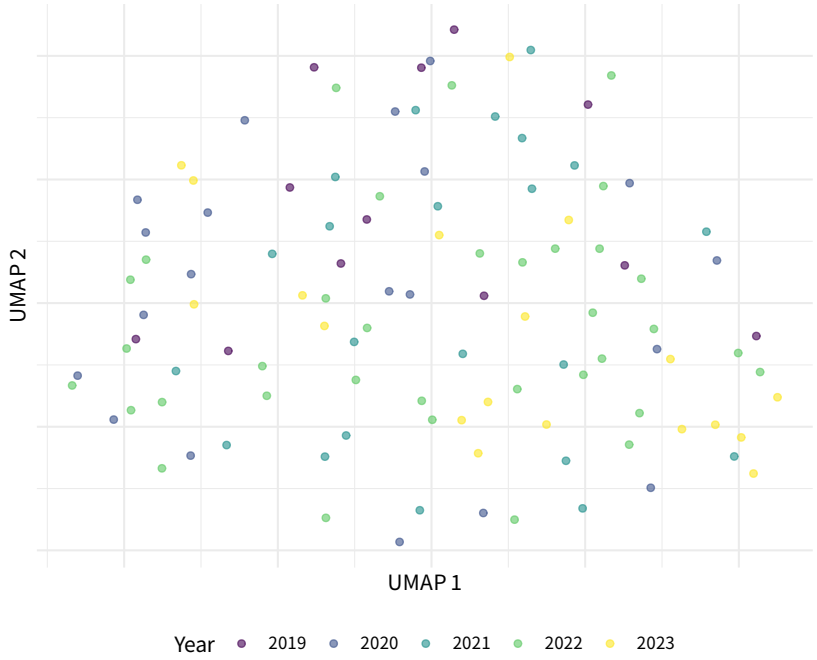


Figure 24.1: UMAP projection of document embeddings, coloured by publication year.

```
cat("Most similar documents:\n")
```

```
#> Most similar documents:
```

```
walk(top_similar, \(i) {
  cat(glue("  [{round(sims[i], 3)}]
   ↪  {text_df_valid$title[i]}\n"))
})
```

```
#> [0.937] Empirical evidence on the relationship between
 ↪  research and teaching in academia[0.931] Relations between
 ↪  academic reputation and innovation networks[0.907] Academic
 ↪  co-authorship is a risky game[0.905] Comments are
 ↪  welcome[0.904] Measuring academic entities' impact by
 ↪  content-based citation analysis in a heterogeneous academic
 ↪  network
```

24.4.6 Embedding clusters vs. citation impact

```
ggplot(text_df_valid, aes(x = umap_x, y = umap_y,
                          colour = log1p(cited_by_count))) +
  geom_point(alpha = 0.6, size = 1.5) +
  scale_colour_viridis_c(option = "C", name = "log(cites + 1)") +
  labs(x = "UMAP 1", y = "UMAP 2") +
  theme_sci() +
  theme(axis.text = element_blank(), axis.ticks =
   ↪  element_blank())
```

24.5 Diagnostics and interpretation

- **Vocabulary coverage**: Check what fraction of abstract words appear in the embedding vocabulary. Low coverage (< 80%) indicates the training corpus is too small or preprocessing is too aggressive.
- **Nearest neighbours**: Inspect the nearest neighbours of key domain terms. If "citation" is closest to "references" and "bibliometric", the embeddings capture domain structure.
- **UMAP stability**: UMAP is stochastic. Run with multiple seeds and check whether the global cluster structure is stable. Individual point positions will vary but clusters should persist.
- **Embedding dimensionality**: 100–300 dimensions is standard. Below 50, embeddings lose semantic resolution; above 300, training becomes slow with diminishing returns.

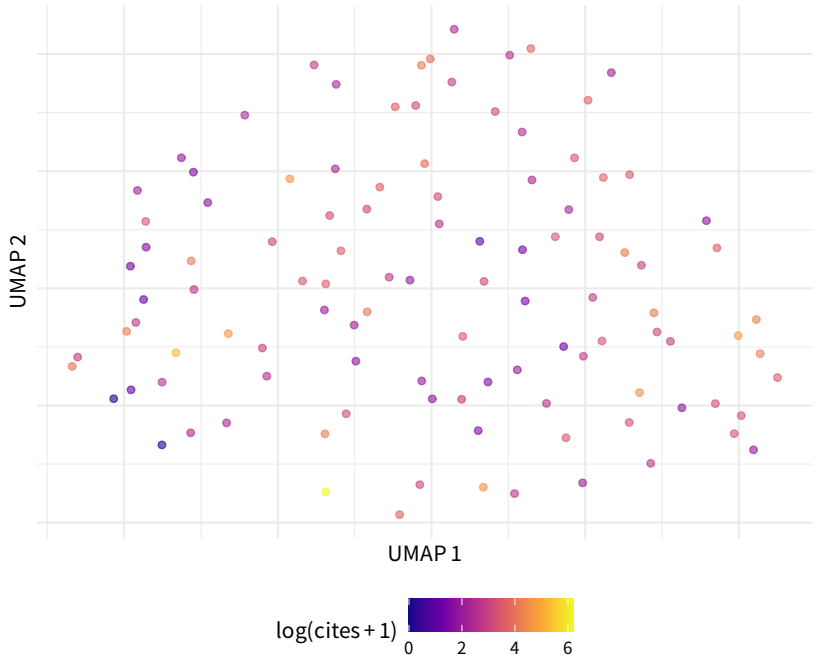


Figure 24.2: UMAP embedding coloured by citation impact (log scale).

24.6 Limitations and responsible use

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- **Embeddings are opaque.** Unlike TF-IDF weights, embedding dimensions have no interpretable meaning. You can measure similarity but not explain why two documents are similar without additional analysis.
- **Training data bias.** Embeddings reflect the biases of their training corpus. If the corpus overrepresents certain topics or perspectives, the embeddings will too.
- **Small corpora produce poor embeddings.** Word2vec needs tens of thousands of documents for reliable training. For small bibliometric corpora (< 1,000 papers), use pre-trained embeddings instead.
- **UMAP/t-SNE distort distances.** These methods preserve local structure but distort global distances. Documents that appear close in 2D may not be the most similar in high-dimensional space (Hicks et al., 2015).

24.8 Common pitfalls

24.9 Common pitfalls

- **Training on too few documents.** Word2vec on 200 abstracts produces unreliable embeddings. Use at least 5,000 documents or switch to pre-trained models.
- **Averaging without weighting.** Simple mean pooling treats every word equally. TF-IDF-weighted averaging gives more weight to distinctive terms and often produces better document embeddings.
- **Over-interpreting UMAP clusters.** Visual clusters in UMAP may reflect projection artifacts. Validate clusters with a separate method (e.g., k-means in the full embedding space).
- **Comparing embeddings from different models.** Embedding spaces are not aligned across different training runs. Cosine similarity is only meaningful within a single embedding space.

24.10 Exercises

1. **Weighted averaging.** Compute document embeddings using TF-IDF-weighted word vectors instead of simple means. Does the UMAP visualisation change?
2. **K-means clustering.** Apply k-means (k = 5, 8, 12) to the document embeddings. Compare the resulting clusters with LDA topics from 22.4. Do they agree?
3. **Pre-trained embeddings.** If available, use SPECTER embeddings (via reticulate and Python) instead of corpus-trained word2vec. How does similarity search quality change?
4. **Analogy test.** Test whether the word2vec model captures analogical relationships (e.g., "journal" - "article" + "book" ≈ "publisher"). What does the result tell you about the model?

24.11 Solutions

Solutions are provided in E.

24.12 Further reading

- Silge and Robinson (2017) — Text embeddings in R, including integration with word2vec.
- Priem et al. (2022) — OpenAlex concept embeddings and their relationship to document content.
- Waltman et al. (2010) — Bibliometric similarity measures as context for embedding-based approaches.

24.13 Session info

```
sessionInfo()
```

```
#> R version 4.6.1 (2026-06-24)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.5 LTS
```

```
#>
#> Matrix products: default
#> BLAS:
 ↪  /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenbl⌋
 ↪  asp-r0.3.26.so;  LAPACK version 3.12.0
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C
 ↪  LC_TIME=C.UTF-8
#>  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8
 ↪  LC_MESSAGES=C.UTF-8
#>  [7] LC_PAPER=C.UTF-8      LC_NAME=C             LC_ADDRESS=C
#> [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8
 ↪  LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods
 ↪  base
#>
#> other attached packages:
#>  [1] uwot_0.2.5               Matrix_1.7-5
#>  [3] word2vec_0.4.1           stm_1.3.8
#>  [5] topicmodels_0.2-17       quanteda.textstats_0.97.2
#>  [7] visNetwork_2.1.4         ggraph_2.2.2
#>  [9] tidygraph_1.3.1          igraph_2.3.4
#> [11] quanteda_4.5.1           pdftools_3.9.1
#> [13] arrow_26.0.0             bibliometrix_5.5.0
#> [15] RefManageR_1.4.0         bib2df_1.1.2.0
#> [17] rcrossref_1.2.1          gt_1.4.0
#> [19] tidytext_0.4.3           glue_1.8.1
#> [21] openalexR_3.1.2          lubridate_1.9.5
#> [23] forcats_1.0.1            stringr_1.6.0
#> [25] dplyr_1.2.1              purrr_1.2.2
#> [27] readr_2.2.0              tidyr_1.3.2
#> [29] tibble_3.3.1             ggplot2_4.0.3
#> [31] tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.3           RColorBrewer_1.1-3
 ↪  rstudioapi_0.19.0
#>   [4] jsonlite_2.0.0         magrittr_2.0.5
 ↪  modeltools_0.2-25
#>   [7] farver_2.1.2           rmarkdown_2.32         fs_2.1.0
#>  [10] vctrs_0.7.3            memoise_2.0.1
 ↪  askpass_1.2.1
```

```
#>  [13] base64enc_0.1-6        htmltools_0.5.9
 ↪  contentanalysis_1.1.1
#>  [16] curl_8.1.0             janeaustenr_1.0.0
 ↪  cellranger_1.1.0
#>  [19] htmlwidgets_1.6.4     tokenizers_0.3.0      plyr_1.8.9
#>  [22] httr2_1.3.0            cachem_1.1.0
 ↪  plotly_4.12.1
#>  [25] dimensionsR_0.0.3      mime_0.13
 ↪  lifecycle_1.0.5
#>  [28] pkgconfig_2.0.3        R6_2.6.1
 ↪  fastmap_1.2.0
#>  [31] shiny_1.14.0           digest_0.6.39
 ↪  patchwork_1.3.2
#>  [34] shinycssloaders_1.1.0  rprojroot_2.1.1
 ↪  RSpectra_0.16-2
#>  [37] SnowballC_0.7.1        labeling_0.4.3
 ↪  urltools_1.7.3.1
#>  [40] timechange_0.4.0       httr_1.4.9
 ↪  polyclip_1.10-7
#>  [43] compiler_4.6.1        here_1.0.2            bit64_4.8.6
#>  [46] withr_3.0.3            S7_0.2.2
 ↪  viridis_0.6.5
#>  [49] ggforce_0.5.0          MASS_7.3-65
 ↪  bibliometrixData_0.3.0
#>  [52] tools_4.6.1            otel_0.2.0
 ↪  stopwords_2.3
#>  [55] zip_3.0.2              rentrez_1.2.4
 ↪  httpuv_1.6.17
#>  [58] promises_1.5.0         grid_4.6.1
 ↪  stringdist_0.9.17
#>  [61] reshape2_1.4.5         generics_0.1.4
 ↪  gtable_0.3.6
#>  [64] tzdb_0.5.0             rscopus_0.9.0          ca_0.72
#>  [67] data.table_1.18.6.1    hms_1.1.4             xml2_1.6.0
#>  [70] utf8_1.2.6             ggrepel_0.9.8
 ↪  pillar_1.11.1
#>  [73] nsyllable_1.0.1       vroom_1.7.1           later_1.4.8
#>  [76] tweenr_2.0.3          lattice_0.22-9        FNN_1.1.4.1
#>  [79] bit_4.6.0              tidyselect_1.2.1       tm_0.7-20
#>  [82] miniUI_0.1.2           knitr_1.52
 ↪  gridExtra_2.3.1
#>  [85] NLP_0.3-3             bookdown_0.48         stats4_4.6.1
#>  [88] crul_1.6.0             xfun_0.62
 ↪  graphlayouts_1.2.5
#>  [91] matrixStats_1.5.0      DT_0.34.0
 ↪  humaniformat_0.6.0
#>  [94] stringi_1.8.9         qpdf_1.4.1            yaml_2.3.12
#>  [97] evaluate_1.0.5         httpcode_0.3.0         cli_3.6.6
#> [100] xtable_1.8-8           systemfonts_1.3.2     Rcpp_1.1.2
```

```
#> [103] readxl_1.6.0           triebeard_0.4.1
 ↪  XML_3.99-0.25
#> [106] parallel_4.6.1         assertthat_0.2.1
 ↪  pubmedR_1.0.2
#> [109] slam_0.1-56            viridisLite_0.4.3
 ↪  scales_1.4.0
#> [112] crayon_1.5.3          openxlsx_4.2.9        rlang_1.3.0
#> [115] fastmatch_1.1-8
```